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MILITARY SPECIFICATION

RADIO SET AN/ARN-21(*)

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 Scope - The equipment covered by this specification when installed in military aircraft shall provide bearing, distance and tone identity information to show the location of the aircraft with respect to a complementary ground beacon (AN/URN-3 or equivalent). This equipment is the airborne portion of the short range air navigation system known as TACAN (Tactical Air Navigation).

1.2 Classification - The Radio Set AN/ARN-21(*) shall consist of the following:

Item	Units	Type Designation	Max. Allow. Size (Inches) W	L	D/H	Max. Allow. Wt. Lbs.	Appl. Para.
1	Receiver-Transmitter Radio	RT-220/ARN-21	11-1/8	18-3/4	8-5/16	63	3.3.1
2	Control, Radio Set	C-1763/ARN-21A	5-3/4	3	4	2.25	3.3.2
3.	Indicator, Range	ID-310/ARN	3-1/4	7-1/4	3-1/4	3.0	3.3.3
4	Indicator, Azimuth	ID-307/ARN	3-1/4	6-5/8	3-1/4	3.0	3.3.4
5	Network, Phase	CV-279/ARN	4-7/8	3	2-3/4	1.0	3.3.5
6	Mounting	MT-928/ARN-21	11-1/8	20	8	7.0	3.3.6

1.3 Accessories - The equipment shall operate with applicable combinations of the following accessories; however, these accessories are not to be supplied as part of this equipment unless otherwise indicated by contract.

Item	Units	Type Designation	Max. Allow. Size (Inches) W	L	D/H	Max. Allow. Wt. Lbs.	Appl. Para.
1	Indicator, Coupler	CU-395/ARN-21A	8-5/8	7-3/8	4-5/16	10.4	3.3.7.1
2	Mounting Base	MT-1530/ARN-21A	7-5/16	8-7/8	1-5/8	0.6	3.3.7.2
3.	Indicator, Bearing Distance-Heading	ID-416/ARN	3-1/4	7-3/8	3-1/4	3.6	3.3.7.3
4.	Indicator, Course	ID-387/ARN	3-1/4	6-5/8	3-1/4	3.0	3.3.7.4
5	Indicator, Course	ID-351A/ARN	3-1/4	7	3-1/4	3.5	3.3.7.5

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6	Indicator, Course	ID-250A/ARN	3-1/4	6-3/8	3-1/4	2.5	3.3.7.6
7	Indicator, Range	ID-388/ARN	3-1/4	5	3-1/4	2.6	3.3.7.7
8	Indicator, Low Approach	ID-525/ARN	3-1/4	6-1/8	3-1/4	1.6	3.3.7.8
9	Indicator, Low Approach	ID-48/ARN	3-1/4	4	3-1/4	1.8	3.3.7.9

(*) The parentheses, in the type designation throughout this specification, shall be replaced by either a number or letter furnished by the procuring agency upon application by the contractor in accordance with the requirements of paragraph 3.2.13 after each contract has been awarded. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Military

MIL-T-152	Treatment, Moisture and Fungus Resistant, of Communications Electronic and Associated Electrical Equipment, General Process for
MIL-E-17555	Electronic Equipment and Associated Maintenance Parts; Preservation, Packaging, Packing and Marking of
MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (For use with Electronic Equipment in Aircraft)
MIL-C-3098	Crystal Units, Quartz
MIL-E-4682	Electron Tubes, Choice and Application Of
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-6781	Control Panel, Aircraft Equipment Rack or Console Mounted
MIL-I-6181	Interference Limits and Tests; Aircraft Electrical and Electronic Equipment
MIL-P-7788	Plate, Plastic, Cockpit and Interior Controls Lighting
MIL-C-5015	Connectors

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MIL-E-7894	Electronic Power, Aircraft Characteristic of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-T-18303	Test Procedures, Preproduction and Inspection; Format for
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment
MIL-I-8692	Range Indicator, ID-310/ARN
MIL-I-18955	Range Indicator, ID-388/ARN
MIL-I-8691	Azimuth Indicator, ID-307/ARN
MIL-I-8693	Indicator Coupler, CU-395/ARN-21A
MIL-N-8689	Network Phase Detecting, CV-279/ARN
MIL-C-18640	Radio Set Control, C-1763/ARN-21A
MIL-C-5824	Indicator, Radio MAG ID-250/ARN
MIL-I-8303	Indicator, Course ID-351/ARN
MIL-I-9229	Indicator, Course ID-387/ARN
MIL-I-25120	Indicator, Low Approach ID-48/ARN
MIL-I-25355	Indicator, Low Approach ID-525/ARN
Exhibit No. USAF WCLN-1117	Indicator, Radio Magnetic, Distance ID-416/ARN

Bureau of Aeronautics

EI-207	Control Panels; Electrical and Electronic Equipment, Rack or Console Mounted, Installation of
EI-224	Installation Instructions for Radio Set AN/ARN-21
SAR-300	Engineering Change Proposals, Electronic Material Changes and Electronic Material Bulletins; Preparation of

STANDARDS

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-287	TACAN Test Signal for Airborne Equipment

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MIL-STD-291 TACAN Standard Signal

DRAWINGS

Air Force - Navy Aeronautical

AND 10461	Connections - Input Power Airborne Electronic Equipment; 28 Volt DC, 115 Volt Single Phase AC, and/or 115/200 Volt 3 Phase AC

2.2 Availability of Documents

(1) When requesting specifications, publications and drawings refer to both title and number. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable specifications and publications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania. Attention: Code ODPT.

(3) Copies of applicable drawings may be obtained upon application to Bureau of Aeronautics, Department of the Navy, Washington 25, D.C., Attention: Avionics Division.

2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring agency.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern.

(1) Parts and Materials as approved by Specification MIL-E-5400 shall be given first consideration.

(2) When the contractors can demonstrate that the use of standard AN or JAN parts or materials will not fulfill the design objectives because of size, weight, performance or other reasons, materials and parts shall be used which most nearly meet or exceed the requirements of the respective standard specification.

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(3) When a non-standard part is used, the meeting of general material requirements such as non-inflammability, fungus resistance, non-toxicity, etc., and, the meeting of environmental conditions as required by specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event that the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass material, electrical and environmental tests. Any assembly built to be expended shall be approved first by the procuring agency.

3.1.1 Non-Standard Part Approval and Material Approval - Approval for the use of non-standard, parts and materials shall be obtained as outlined in Specification MIL-E-5400 except as follows:

(1) Mechanical parts are expected to meet the material requirements of MIL-E-5400, and shall be submitted as non-standard parts only when the material requirements are not met.

(2) Non-standard parts, as defined in Specification MIL-E-5400 of an electrical nature must be approved before the pre-production model of the equipment is delivered to the Bureau of Aeronautics. The request for approval must be forwarded by the contractor before the part is purchased or fabricated. In addition to or in lieu of the NON-Standard Part Data Sheet the contractor is requested to submit itemized lists of the parts, their sources, and the contractor's drawings or purchase specification covering the parts.

3.1.2 Choice of Tube Types - Electronic tubes used in the equipment shall be chosen in accordance with the following requirements and in the sequence noted:

(1) MIL-E-4682

(2) The use of any tube not in accordance with (1) above must be approved by the Bureau of Aeronautics; however, before consideration for approval, the contractor must submit the following information:

(a) Quantitative data on the electrical advantages and mechanical adequacy of the proposed tube over the most nearly equivalent type on the acceptable listings.

(b) Test data and performance limits of the tube which must be maintained to insure satisfactory and reliable performance in the equipment.

(c) Information as to whether the tube manufacturer has received Military or Navy qualifications approval for the particular tube.

3.1.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives to be considered in

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the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the Bureau of Aeronautics for consideration. Each request should be accompanied by complete supporting information.

3.1.4 Resistors - Preference shall be given to the use of ten percent tolerance insulated resistors. Five percent and smaller tolerance resistance shall not be used unless necessary to meet the performance requirements of this specification.

3.1.5 Crystals - All quartz crystals shall comply with the requirements of MIL-C-3098.

3.1.6 Motors - All motors shall have sealed bearings that do not require lubrication or other service for a minimum of 5000 hours of equipment operation.

3.1.7 Connectors - All connectors shall be in accordance with MIL-C-5015.

3.2 General Requirements

3.2.1 General - This equipment shall fulfill all applicable design and performance requirements of Specification MIL-E-5400 for Class I equipment, unless otherwise specified herein.

3.2.2 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment:

Temperature	Room ambient 25°C + 15°C or -5°C
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	115 ± 1.0 V AC and/or 27.5 ± 0.5 V DC

3.2.3 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions or reasonable combinations of service conditions as specified in MIL-E-5400 for Class I equipment.

3.2.4 Total Weight - The total weight of the equipment excluding cables shall not exceed the applicable combinations of weight listed in 1.2.

3.2.5 Operational Stability - The equipment shall operate with optimum performance for 200 hours, continuously or intermittently without necessity for readjustment of any controls which are non-accessible during flight

3.2.6 Operating Life

3.2.6.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 1000 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.6.2 Total Operating Life - The equipment shall have a minimum total operating life of 5000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer, and submitted to the proving agency within six months after delivery of the first production unit.

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3.2.7 Interconnection Wiring - This equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded, when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished.

3.2.8 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with the requirements of Specification MIL-E-5400. However, cables and that portion of the connectors which is to be attached to the cables shall not be supplied as part of the equipment.

3.2.9 Console Controls - All remote control units shall be satisfactory for mounting in a console arrangement when used in aircraft. These units shall meet the requirements of Specification MIL-C-6781, shall have a plastic lighting plate meeting the requirements of Specification MIL-P-7788 and shall be equipped with fasteners meeting the requirements of Specification MIL-F-25173 for mounting in accordance with requirements of Specification EI-207. The lamps shall be connected in parallel with one side grounded to the chassis of the unit and the other side being connected to a separate contact, in the electrical connector receptacle of the unit, for connection to an external +28 V input lead. The brilliance of the lamps will be controlled externally and the brilliance control shall not be part of the control unit. The arrangement of controls and all lettering and markings proposed for use on the front panel of control units shall be approved by the Bureau of Aeronautics prior to fabrication of any complete units.

3.2.10 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance from the following power sources with characteristics as defined in MIL-E-7894 having limits as specified herein. The power required shall not exceed the specified amounts.

(1) AC Power (Single Phase)

(a) 115 Volt Variable Frequency

Operating Voltage Limits	102 to 124 Volts
Operating Frequency Limits	320 to 1000 cps
Power (1Ø)	370 V. A.

(b) 115 Volt Constant Frequency

Operating Voltage Limits	102 to 124 Volts
Operating Frequency Limits	380 to 420 cps
Power (1Ø)	75 V. A.

(2) DC Power (28 Volt System)

Operating Voltage Limits	24 to 29 Volts
Power (Maximum Steady State)	50 Watts

3.2.10.1 Equipment Power Factor - The equipment shall not present a power factor less than .85 lagging or less than .95 leading for the 320-1000 cps source or 75 lagging or .95 leading for the 380 to 420 cps source.

3.2.10.2 Input Power Connection - Input power connections shall be in accordance with drawing AND 10461.

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3.2.10.3 Low Frequency Damage - Equipment damage shall not result from the application of 115 volts, 260 cycles, at room temperature for a period of 30 minutes, to those circuits normally operating from 320 to 1000 cycles. In addition, equipment damage shall not result from the application of 115 volts, 360 cycles, at room temperature for a period of 30 minutes, to those circuits normally operating from 380 to 420 cycles. The equipment shall automatically resume operation when the frequency returns within limits.

3.2.11 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

3.2.11.1 Under Voltage Protection - The equipment shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within limits.

3.2.12 Moisture and Fungus Resistant Treatment - Equipment shall be fungus proofed by selection of parts and materials as non-nutrient for fungus, or the parts and materials shall be treated prior to their use in the equipment so that overall spraying of the equipment is not necessary. Overall spraying of the equipment in accordance with JAN-T-152 shall be made in the event selection of parts and materials described above is not possible.

3.2.13 Nomenclature and Nameplates - Nomenclature assignment and Nameplate approval for equipment identification shall be in accordance with the requirements of Specification MIL-N-18307.

3.2.14 Interchangeability - The interchangeability requirements between like assemblies, subassemblies and replaceable parts as specified in MIL-E-5400 shall be extended to require interchangeability of units of this equipment with the latest model of any previously produced equipment. Such interchangeability shall be measured against a model, manufacturing drawings or other technical information as provided by the Bureau of Aeronautics for the purpose. In the event that requirements of MIL-E-5400 conflict with the requirements for interchangeability, the interchangeability requirements shall govern.

3.2.15 Provisions for Maintenance and Field Testing - The equipment shall provide for ease of maintenance to the greatest extent practical. Provisions for field testing shall be as specified in MIL-E-5400.

3.2.16 Radio Interference Control

3.2.16.1 On Frequency - With the equipment tracking on a signal of -80 dbm, the specified performance shall not be affected by the following types and levels of interference at any frequency on or within 3 megacycles of the desired frequency:

	Type	Level
1.	6000 randomly spaced single pulses per second each 3.5 microseconds wide.	-20 dbm
2.	CW	-85 dbm

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3.2.16.2 Off Frequency - The noise level at the output of the second detector with no desired signal present shall not be changed by CW signals of the following frequencies and amplitudes at the antenna terminal:

Frequency	Amplitude	
0.1 mc to 950 mc	-30 dbm	
63 mc (IF)	-10 dbm	
Image Frequency	-30 dbm	
1250 to 10000 mc	-30 dbm	With SIG applied through an appropriate filter

3.2.16.3 Radiated and Conducted Interference Requirements - The equipment shall meet the radiated and conducted interference and susceptibility requirements of MIL-I-6181. The equipment shall comply with these requirements in all normal modes (i.e. search, track, transmit-receive and receive) of operation and also during channel selection. In addition at all frequencies between 1250 mc and 10000 mc, the spurious antenna conducted output of the transmitter under transmit conditions shall not be more than -17 dbm. At frequencies between 950 mc and 1250 mc the spurious antenna conducted output of the transmitter under transmit conditions shall not be more than 1 milliwatt. The method of measurement between 1,000 mc and 10,000 mc shall be approved by the procuring agency.

3.2.16.4 Antenna Conducted CW - The antenna CW conducted power from the equipment between pulses or in the receive mode of operation in the frequency range of 1015 to 1045 mc shall not be greater than a level 107 db below the peak of the transmitter pulse.

3.2.17 Performance -

3.2.17.1 Operation - A working assembly of the equipment when properly interconnected using standard accessories shall transmit to and receive from a complementary navigation beacon a series of pulses from which it shall derive information to perform the following functions:

(1) Determine the radial to the navigation beacon on which the aircraft is flying and actuate the Azimuth indicator to continuously display the bearing to the beacon.

(2) Determine the distance of the aircraft from the navigation beacon and actuate the range indicator to continuously indicate the distance to the beacon.

(3) Determine the deviation of the aircraft from the selected course or beacon radial and actuate the vertical needle of the diviation indicator.

(4) Determine the identity of the beacon to which the equipment is channeled by aural indication.

(5) Provide a blanking pulse output during the transmit condition which will cause blanking of other equipment operating in the same frequency band.

3.2.17.1.1 Characteristics of Tacan Signals - This equipment shall operate with pulse code and pulse amplitude type signals of a ground facility meeting the requirements of MIL-STD-291.

3.2.17.1.2 Warm-up Time - The equipment shall meet all performance requirements after a maximum warm up time of 5 minutes.

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3.2.17.2 Remote Control - Complete operational remote control of the receiver-transmitter shall be through the control panel.

3.2.17.3 Operational Range - The operational range of the equipment shall be such that reliable bearing, distance and identity information will be maintained from zero out to a line of sight distance of 195 nautical miles from the beacon. Under no circumstances shall the equipment lock on its own transmission. No degradation of any of the other functions of the equipment shall occur as a result of its own transmission.

3.2.17.4 Accuracy - The accuracies specified below shall be determined using a standard TACAN test signal as covered by MIL-STD-287 and shall be realizable using any combination of applicable aircraft instruments and accessories.

3.2.17.4.1 Distance Accuracy (DME) - The equipment shall have a distance accuracy of ± (0.1 mile plus 0.2% of the distance) under both static and dynamic tracking conditions. This accuracy shall be maintained over a range of signal strength from -90 dbm to 0 dbm. When operating with a beacon which is only 70% efficient in replying to interrogations, the equipment shall be capable of locking on and continuously tracking distance replies, while moving at speeds up to 2,000 knots in either direction with respect to its own interrogation.

3.2.17.4.2 Bearing Accuracy - The equipment shall have bearing accuracies as shown in the table below under both static and dynamic tracking conditions. The equipment shall be capable of locking on and continuously tracking bearings changing in either direction at rates up to five degrees per second over a full 360° range.

Bearing	Bearing Accuracy	Signal Strength
All bearings	± 2°	-90 dbm to -84 dbm
Other than 90°	± 1°	-84 dbm to 0 dbm
90°	± 2°	-86 dbm to -80 dbm
	± 1°	-80 dbm to 0 dbm

3.2.17.4.2.1 15 Cycle Reference Bearing Pulse - The phase relationship between the 15 cycle reference pulse and the 135 cycle component of the modulation envelope, in passing through the equipment, shall be held within ± 5° of the correct phase relationship at all bearings. This requirement shall be applicable over a 5% to 55% range of modulation.

3.2.17.5 Beacon Identification - The equipment shall provide intelligent and unambiguous aural identification of the selected beacon signal with a 10 db signal plus noise to noise ratio within the range of signal strength of -90 dbm to 0 dbm. The beacon shall be identifiable by a Morse coded audio tone of 1020 cycles per second. The audio output shall be adjustable to provide 0 to 100 milliwatts output into a 600 ohm resistive load. Accidental shorting or opening of the output circuit external to the equipment shall not cause damage to the equipment or render other functions of the equipment inoperative.

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3.2.17.6 Sensitivity - The sensitivity of the equipment shall be -90 dbm or better. Sensitivity is defined as that signal level necessary to enable the equipment to meet all the performance requirements of this specification with respect to the distance, bearing and tone identity functions of the equipment.

3.2.17.7 Selectivity -

3.2.17.7.1 Adjacent and Other Channels - The equipment shall have greater than 50 db rejection to any normal beacon signal or signals other than the channel to which the equipment is tuned. This requirement applies both in the presence or absence of a desired beacon signal on the selected frequency channel. In the presence of a desired signal, the undesired signal shall be considered rejected if all the performance requirements of this specification are met with the exception of allowing an additional 1° bearing error. In the absence of a desired signal, the undesired signal shall be considered rejected if no distance, bearing or identity tone information is obtained.

3.2.17.7.2 Co-Channel - The equipment shall be capable of rejecting the weaker of two beacon signals received on the same channel when the difference in signal strength is 8 db or more. The performance obtained from the stronger signal shall meet all requirements of this specification except that an additional 1° bearing error is permitted, when only 8 db of difference exists between the two signals.

3.2.17.8 Automatic Gain Control - When the input signal is varied by as much as 64 db the automatic gain control circuits shall act to maintain the bearing and distance outputs within the accuracy and performance requirements of this specification.

3.2.17.9 Decoding - The sensitivity of the receiver shall not decrease more than 3 db when receiving a pulse pair spaced ± 0.5 microsecond from nominal, and loss of receiver sensitivity shall be at least 30 db for a pulse pair spacing ± 4.0 microseconds for nominal.

3.2.17.10 Frequency Channeling - 126 crystal controlled frequency channels shall be provided at one megacycle spacing. Channels 1 through 63 shall pair receiver frequencies 962 through 1024 megacycles with transmitter frequencies 1025 through 1087 megacycles and channels 64 through 126 shall pair receiver frequencies 1151 through 1213 megacycles with transmitter frequencies 1088 megacycles through 1150 megacycles.

3.2.17.11 Channel Selection Time - The time required by the equipment to tune from one channel to any other channel shall not exceed 12 seconds.

3.2.17.12 Blanking Pulse Output - The equipment shall be equipped with a blanking pulse output with the following characteristics:

(1) Positive Polarity

(2) 15 to 70 volts amplitude across 2200 ohms ± 10% in parallel with 500 micromicrofarads load impedance.

(3) 20 volts per microsecond minimum rise time.

(4) One to five microsecond duration.

(5) 10 volts per microsecond minimum decay time. (90% to 10% amplitude)

(6) One pulse shall be generated for each interrogation out of the equipment and the leading edge of this pulse shall occur at least 0.8 but not more than 5 microseconds before the 10% amplitude point of the leading edge of the first pulse of the interrogation pulse pair.

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(7) Spurious output associated with the blanking pulse shall be below 1 volt in amplitude

(8) A blanking pulse shall be generated only when the equipment is in the transmit condition of operation.

3.2.17.13 Calibration - The equipment shall read zero miles ± 0.1 mile when the delay between the 50% amplitude points of the second rf interrogation pulse and the second rf reply pluse is 50 microseconds.

3.2.17.14 Peak Power Output - The peak power of each pulse in a pulse pair transmitted by the equipment in T/R mode operation during the search condition shall be at least 1.5 KW on all channels. The two pulses in a pulse pair shall not differ in power by more than 1 db. The above power measurements shall be determined using a 50 ohm load impedance. Not less than 55% of full transmitter output power shall be delivered to a load presenting a standing wave ratio of 2 to 1 at the worst phase angle.

3.2.17.14.1 Altitude Power Reduction - If necessary to enable compliance with the temperature and altitude conditions of MIL-E-5400 at altitudes above 20,000 feet, it will be permissible to reduce the power output of the equipment by not more than 3 db by means of a barometric switch.

3.2.17.15 Modulation Characteristics -

(1) Pulse Shape - The pulse envelope as detected by a linear detector shall have a rounded shape falling within the following limits.

(2) Pulse Top - The amplitude of each pulse between the 90% amplitude points shall be flat to within ± 1 db.

(3) Pulse Duration - The pulse duration measured between 50% amplitude points shall be 3.5 ± 0.5 microseconds.

(4) Pulse Coding - The pulse code spacing measured between 50% amplitude points from leading edge to leading edge shall be 12 ± 0.5 microseconds.

(5) Pulse Repetition Rate - The pulse repetition rate shall fall between 120 and 150 pulse pairs per second in the track condition. The pulse repetition rate shall vary or jitter enough to preclude synchronization of transmitted pulses between any of 100 equipments on the same channel for more than one second in one hour.

3.2.17.16 Transmitted Spectrum - In both the search and track conditions, 90% of the transmitted energy shall fall within a 5 mc bandwidth centered on the nominal channel frequency of that channel, and the remaining energy shall decay smoothly with frequency.

3.2.17.17 Speed of Response -

(1) Range Search Time - A complete range cycle shall not exceed 30 seconds, with and without a signal applied.

(2) Bearing Search Time - A complete bearing search cycle shall not exceed 20 seconds, with and without a signal applied.

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(3) Range Memory - Upon loss of signal the memory period during which the equipment retains its range reading shall fall between 8 and 12 seconds.

(4) Bearing Memory - Upon loss of signal, the memory period during which the equipment retains its bearing reading shall fall between 1 and 3 seconds.

3.2.17.18 Adjustment Controls - Adjustment controls necessary for calibrating the equipment when installed within the aircraft shall be accessible through the front panel.

3.2.17.19 Workmanship - Workmanship , fabrication and manufacturing processes shall conform to the requirements of MIL-E-5400.

3.3 Detail Requirements -

3.3.1 Receiver-Transmitter, RT-220/ARN-21 - The RT-220/ARN-21 shall be the receiver-transmitter portion of the equipment.

3.3.1.1 Stability

3.3.1.1.1 Transmitter and Local Oscillator Stability - Each of the receiver-transmitter frequency channels shall be maintained within ± 100 kilocycles of its assigned frequency. Up to a two to one standing wave ratio at the transmitter with a ten to thirty foot length of RG-8/U cable between the transmitter and load or the injection of up to 0.1 volt of CW at any frequency into the antenna terminal shall not cause the transmitter frequency to fall outside the specified limits.

3.3.1.1.2 IF Stability - The IF center frequency of the receiver-transmitter shall be maintained within ±50kc of 63 mc. If an IF frequency of other than 63 mc is employed the stability shall be within ± 0.08% of the center frequency.

3.3.1.2 Form Factor - The receiver-transmitter shall be completely housed in a case. The front panel shall be flanged back on the case to provide electrical bonding and mechanical strength. The receiver-transmitter shall be secured within its case by means of quick disconnect type fasteners. The handles or other panel mounted fixtures shall not extend more than two inches beyond the panel face. Outline dimensions and form factor shall conform to drawing EI-224-2.

3.3.1.3 Connections - A single multi-contact connector shall be mounted on the lower portion of the rear face of the receiver-transmitter chassis so as to automatically effect all power, control and output circuit connections to the mounting base when the receiver-transmitter is mounted. An aperture shall be provided in the receiver-transmitter case to permit engagement of this connection with the mating connector which shall be mounted at the rear of the Mounting Base MT-928/ARN-21. One HN type receptacle for the antenna connection shall be provided on the front panel for both reception and transmission. In addition a suitable type connector shall be provided for the blanking pulse output.

3.3.1.4 Receiver- The receiver shall employ a preselector, converter, intermediate frequency amplifier and suitable detecting, decoding and bearing and distance measuring circuits with output circuits to operate the bearing, distance and deviation indicators.

3.3.1.4.1 Input - The r-f input shall connect through the HN type antenna receptacle to a 50 ohm coaxial line. (The standing wave ratio shall be less than two to one over the specified frequency range. The receiver shall not be damaged by the transmitter under any conditions of mismatch. Operating the equipment with the cable off or the plug accidently shorted shall not damage the receiver transmitter.)

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3.3.1.4.2 Preselector - The preselector shall be tunable over the full receiving band of the equipment by means of the electric tuner in the receiver-transmitter.

3.3.1.4.3 Local Oscillator - The local oscillator energy shall be supplied from the transmitter oscillator.

3.3.1.4.4 Decoders - Suitable decoders shall be provided to discriminate against improperly coded pulses and to separate the reference and bearing signals for use in the phase comparison and tracking circuits.

3.3.1.4.5 Bearing Measuring Circuits - The bearing measuring circuits shall be designed to automatically determine the reciprocal of the bearing of the equipment from the navigation beacon by measuring the phase shift in degrees required to properly phase the decoded beacon bearing signals with the decoded beacon reference signals.

3.3.1.4.6 Resolvers - The equipment shall employ the resolvers in the course indicator as phase shifters for the phase detector. A servo amplifier shall be provided in the receiver to drive the motor and resolvers in the indicator to shift the phase of the bearing signals until they are properly phased with the reference signals in the phase detector.

3.3.1.4.7 Distance Measuring Circuit - The distance measuring circuits shall be designed to automatically determine the distance in nautical miles of the equipment from the navigation beacon by measuring the time delay between transmitted interrogation pulses and properly coded beacon reply pulses.

3.3.1.4.8 Strobe - The search and tracking strobe shall be of the electromechanical type. The equipment shall employ the linear shaft output in the range indicator to balance the servo loop. A servo amplifier shall be provided in the receiver to drive the motor and linear shaft to properly balance the servo loop.

3.3.1.4.9 Identification Circuits - Receiver circuits shall be provided to translate the mark beacon identification pulses into audio tones in an output circuit having suitable impedances.

3.3.1.5 Transmitter - The transmitter shall employ a crystal oscillator, a multiplier, a power amplifier and a modulator to produce coded pulses to interrogate the navigation beacon.

3.3.1.5.1 Modulator - The modulator shall be controlled by the search and tracking circuits but shall be designed for additional external coding when required. The modulator shall be of the hard tube suitable for high duty cycle operation.

3.3.1.5.2 Oscillator - The oscillator shall be crystal controlled. The oscillator frequency shall be selected by means of the electronic tuner.

3.3.1.5.3 Power Amplifier - The power amplifier shall be tunable over the full transmitting band of the equipment by means of electronic tuning. It shall amplify the desired frequencies provided by the crystal oscillator through suitable multiplying circuits.

3.3.1.5.4 Output - The r-f output shall be connected through the HN type receptacle to a 50 ohm coaxial line.

3.3.1.6 Power Supply - The d-c power supply shall be of a high capacity electronically regulated type capable of supplying the receiver-transmitter with the proper rectified voltages from the 115 volt 320 to 1000 cycle AC source. All filament voltages and other AC voltages derived from the 320 to 1000 cycle AC source shall be regulated.

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3.3.2 Control, Radio Set, C-1763/ARN-21A - The Radio Set Control, C-1763/ARN-21A shall provide for control of power applied to the RT-220/ARN-21 and the desired channel of operation is selected. Refer to specification MIL-C-18640 for detailed requirements.

3.3.3 Indicator, Range, ID-310/ARN - Range Indicator ID-310/ARN is a distance indicator which displays the distance from an aircraft to a navigation beacon when connected directly to a Receiver-Transmitter RT-220/ARN-21. No Indicator Coupler CU-395/ARN-21A is required for this indicator. Refer to specification MIL-I-8692 for detailed requirements

3.3.4 Indicator, Azimuth ID-307/ARN - Indicator, Azimuth ID-307/ARN displays the bearing of an aircraft to a navigation beacon when connected to a Receiver-Transmitter RT-220/ARN-21. Refer to specification MIL-I-8691 for detailed requirements.

3.3.5 Network, Phase Detecting, CV-279/ARN - Network, Phase Detecting CV-279/ARN combines input from (1) the fundamental resolver of an ID-307/ARN Azimuth Indicator and (2) the course selector synchro resolver of an ID-249A/ARN Course Indicator. The unit operates the vertical pointers, ambiguity meters of two deviation indicators as well as an automatic pilot. Refer to specification MIL-N-8689 for detailed requirements.

3.3.6 Mounting MT-928/ARN-21 - Mounting MT-928/ARN-21 shall mount the RT-220/ARN-21 receiver-transmitter in the aircraft. Refer to EI-224 for outline drawings. The base, a vibration isolator type, shall meet the applicable requirements of specification MIL-C-172 and shall be equipped with a suitable mechanical extractor unit for removing the equipment from the aircraft.

3.3.6.1 Shock Mounting - A vibration and shock absorbing system shall be provided in accordance with MIL-C-172 and shall be capable of meeting shock and vibration requirements of Specification MIL-E-5400, Class I, equipment.

3.3.6.2 Connections - A single multi-contact connector shall be mounted at the rear of the mounting base to effect all power, input and output circuit connections to the receiver-transmitter when the receiver-transmitter is mounted on and attached to the base.

3.3.7 Accessories - The equipment shall operate with the following accessories, however, these accessories shall not be supplied as part of the equipment.

3.3.7.1 Indicator Coupler, CU-395/ARN-21A - The Indicator Coupler, CU-395/ARN-21A is the unit used to couple the outputs of Receiver-Transmitter RT-220/ARN-21 to flight instruments for presentation of bearing and distance of an aircraft to a navigation beacon. Refer to specification MIL-I-8693 for detailed requirements

3.3.7.2 Mounting MT-1530/ARN-21A - Mounting MT-1530/ARN-21A is a vibration isolator type mount provided to support the CU-395/ARN-21A Indicator Coupler Unit. Refer to EI-224 for outline drawings. The mounting base shall meet the applicable requirements of specification MIL-C-172.

3.3.7.3 Indicator, Bearing-Distance-Heading ID-416(XA-)/ARN - Indicator, Bearing-Distance-Heading displays magnetic heading (or ground tracks), also, course and distance to a reference destination from the aircraft when connected to a receiver-transmitter RT-220/ARN-21 through indicator coupler CU-395/ARN-21A. Refer to Air Force Exhibit No. WCLN-1117 for detail requirements.

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3.3.7.4 Indicator, Course ID-387/ARN - Indicator, Course ID-387/ARN provides the facilities of a deviation indicator, flags to indicate non-operation of associated equipment, a relative heading indicator, a course selector, an ambiguity indicator, and a marker beacon indicator. Refer to specification MIL-I-9229 for detailed requirements.

3.3.7.5 Indicator, Course ID-351A/ARN - Indicator, Course ID-351A/ARN contains the same facilities as the ID-387/ARN except that installation requirements differ to a small extent. Refer to specification MIL-I-8303 for detailed requirements.

3.3.7.6 Indicator, Course ID-250A/ARN - Indicator, Course ID-250A/ARN displays magnetic heading on a compass dial and magnetic bearing of two radio stations under two pointers when connected to receiver-transmitter RT-220/ARN-21.
Refer to specification MIL-C-5824A for detailed requirements.

3.3.7.7 Indicator, Range ID-388/ARN - Indicator, Range ID-388/ARN displays distance of an aircraft to a navigation beacon when connected to a receiver-transmitter RT-220/ARN-21 through Indicator-Coupler CU-395/ARN-21A. Refer to specification MIL-I-18955 for detailed requirements.

3.3.7.8 Indicator, Low Approach ID-525/ARN - Indicator, Low Approach ID-525/ARN provides the facilities of a cross pointer with associated flags to indicate non-operation of associated equipment and a marker beacon lamp. Refer to specification MIL-I-25355 for detailed requirements.

3.3.7.9 Indicator, Low Approach ID-48/ARN - Same as ID-525/ARN except that there is no associated marker beacon light on the instrument. Refer to specification MIL-I-25120 for detailed requirements.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests

(1) Preproduction Tests - Preproduction Tests are those tests conducted on an equipment representative of the equipments to be supplied under the contract, for the purpose of determining that the equipment meets all the requirements of the applicable specifications. Preproduction Tests shall be accomplished by Contractor Demonstration Tests and Service Approval Tests.

(a) Contractor Demonstration Tests - Contractor Demonstration Tests shall be accomplished under the responsibility of the contractor to demonstrate that the equipment complies with all applicable requirements.

(b) Service Approval Tests - Service Approval Tests shall be accomplished under the responsibility of the procuring service for the purpose of verifying the results of the Contractors Demonstration Tests and for conducting any additional tests deemed necessary to determine that the equipment complies with all applicable requirements.

(2) Acceptance Tests - Acceptance Tests are those tests accomplished on equipment submitted for acceptance under the contract. Acceptance Tests shall consist of the following:

(a) Individual Tests - Individual Tests are those tests conducted on each equipment submitted for acceptance under the contract.

(b) Sampling Tests - Sampling Tests are those tests accomplished on a quantity of equipments as approved by the Bureau of Aeronautics and which have previously passed Individual Tests.

(c) Special Tests - Special Tests are those tests accomplished on a quantity of equipments as approved by the Bureau of Aeronautics.

(3) Life Tests - Life Tests shall be conducted by the contractor at his plant. These tests shall be started as soon as practicable on an early production equipment.

4.2 Preproduction Tests -

4.2.1 Contractor Demonstration Tests - The contractor shall conduct tests on one or more equipments as necessary to determine that the model as proposed by the contractor will meet all the requirements of the applicable specifications and the contract. Contractor Demonstration Tests shall be conducted in accordance with the approved test procedure. The data obtained by the contractor in conducting these tests shall be included with the design data submitted with the equipment for Service Approval Tests. The Inspector of Naval Material and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. When tests are supervised the Bureau of Aeronautics may, when conducting Service Approval Tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting Service Approval Tests, the Bureau of Aeronautics reserves the right to repeat any test previously conducted when deemed necessary. Contractors not having laboratory facilities to conduct all required tests shall obtain either the services of a commercial testing laboratory or written approval from the Bureau of Aeronautics to omit that portion of the tests which cannot be conduct

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4.2.2 Service Approval Tests - The Service Approval Sample shall consist of one equipment representative of the production equipment to be supplied under the contract. This equipment shall be delivered by the contractor to the destination specified by the Bureau of Aeronautics.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for Service Approval Tests, the contractor shall also submit the accessory material and design and test data specified in MIL-E-5400. The design and test data shall indicate the physical and electrical characteristics of the equipment and establish that the equipment meets all applicable requirements. The test data shall include data obtained by the contractor in testing to MIL-T-5422.

4.2.3 Scope of Tests - Preproduction Tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction Tests shall include environmental tests in accordance with MIL-T-5422 and interference tests in accordance with MIL-I-6181. The sampling and testing of plastic lighting plates shall be in accordance with MIL-P-7788.

4.2.4 Preproduction Approval - Approval of the preproduction sample shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No additional equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of any equipments prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance under the contract. The preproduction samples shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the required Acceptance Tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Bureau of Aeronautics shall engage the services of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all Acceptance Tests required by this specification and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Acceptance Tests shall consist of the following:

(1) Individual Tests - Individual Tests shall be conducted on each equipment submitted for acceptance under the contract.

(2) Sampling Tests - Sampling Tests shall be conducted on a quantity of equipments as specified in 4.3.2.1. Equipments for sampling tests shall be selected by the Government Inspector after the equipment has passed the Individual Tests and has been packaged and packed ready for shipment.

(3) Special Tests - Special Tests shall be conducted on a quantity of equipment as approved or requested by the Bureau of Aeronautics.

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4.3.1 Individual Test - Each equipment submitted for acceptance shall be subjected to the Individual Test to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum each equipment accepted must successfully pass the tests specified in the following paragraph.

4.3.1.1 Scope of Test - As a minimum, each equipment shall be subjected to the following:

(1) Examination of Product

(2) Operational Test

(3) Reliability Test

4.3.1.1.1 Examination of Product - Each equipment shall be carefully examined to determine that the requirements for material and workmanship as specified in MIL-E-5400 has been met. Also, it shall be determined that approval has been granted for the use of all non-standard parts.

4.3.1.1.2 Operational Test - Each equipment shall be operated. The period of operation shall be long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.1.3 Reliability Test - Each equipment shall be operated under the conditions specified below for a period of 12 hours without a failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only such adjustments shall be permitted which can be made using such controls and adjustments which would normally be accessible during flight.

Temperature	Ambient Room
Humidity	Ambient Room
Vibration (15 minute) (Without Isolators)	Any frequency within the range of 20 to 60 cps and an amplitude of ± 5 g's

The fifteen (15) minute vibration run shall be made at the beginning of the 12 hour period. The equipment shall be operating during the vibration run. During the twelve (12) hour period of operation the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur it shall be repaired and the test started over, except the fifteen (15) minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action must be taken to eliminate this defect from future equipment. A record shall be kept of all failures.

4.3.2 Sampling Test - Sampling Tests shall be conducted on a quantity of equipment as specified in the following paragraph. Equipment selected must first have passed all of the individual tests. Sampling Tests shall be conducted on complete equipments.

4.3.2.1 Sampling Test Schedule - Sampling Tests shall be performed on the following quantity of equipment. Selection of equipment for tests shall be under the supervision of the naval inspector.

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Quantity Delivered for Acceptance	Quantity to be tested
From the first 10	1
From 11 to 100	2
From 101 to 200	1
From 200 and above	1 for each additional 200 or fraction thereof

4.3.2.2. Scope of Test - As a minimum each equipment selected for Sampling Test shall be subject to the tests specified below.

Tests	
I	Conduct a complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure all applicable specification requirements have been met.
II	Repeat the above using the highest and lowest specified input voltages and frequencies.
III	Conduct the reliability test specified in paragraph 4.3.1.1.3 except the test duration shall be 50 hours and two failures shall be permitted.

4.3.3 Special Test - Special Tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. Equipment selected for Special Test may be selected from equipment previously subjected to the Sampling Test.

4.3.3.1 Special Test Schedule - In selecting equipment for Special Tests the following schedule shall be followed:

(1) An early production equipment.

(2) An equipment midway of the production run.

(3) An early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional tests are necessary.

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory quality control.

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4.3.3.2 Scope of Test - In conducting Special Tests, test procedures previously approved for the Preproduction and Sampling Tests shall be used where applicable. When not applicable the contractor shall prepare a test procedure and submit to the Bureau of Aeronautics for approval prior to conducting the tests. As a minimum the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports only such tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (90 minutes cycling - 30 min. in each of 3 planes— Freq 5 to 500 cps ± 5g)

(3) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the Sampling or Special Test, the following corrective action shall be taken.

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) If a design defect, suggest corrective action for Bureau of Aeronautics approval.

(4) Where practical, include a test in the Individual Test to check all equipments for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Life Tests - The equipment shall be subjected to a Life Test at the contractor's plant under the following simulated service conditions.

Temperature	Normal room temperature
Altitude	Normal (0 to 5000 ft.)
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. Voltage	26.5 ± 2.5 volts

4.4.1 Test Data - The contractor shall keep a daily record as to the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures during this test, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part Type number

(2) The circuit reference symbol-number

(3) The part function

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(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

In the event of a part failure the naval inspector shall be notified immediately. A report shall be submitted to the Bureau of Aeronautics upon completion of the test. In this report the contractor shall propose suitable and adequate design or material corrections for all failures which occured. The Bureau of Aeronautics will review such proposals and determine whether they are acceptable.

4.4.2 Test Duration - The duration of the Life Test shall be 300 hours.

4.4.2.1 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically the equipment shall be turned on and off several times and put through all phases of its performance.

4.4.2.2 Performance Check - At approximately eight hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the Bureau of Aeronautics.

4.4.3 Life Test Schedule - The equipment to be tested shall be representative of the current production in all significant aspects of design and manufacture. It shall be selected by the Government Inspector from equipments which have passed the Individual Tests. The Life Test shall be performed on the following quantities of equipment unless otherwise stated in the contract or approved by the Bureau of Aeronautics.

Quantity Delivered for Acceptance	Quantity to be tested
First 25	1
Next 26 to 200	1
Next 201 to 500	1
Over 500	1 for each additional 500 or fraction thereof

4.4.4 Reconditioning of Test Samples - Equipment which has been subjected to the Life Test shall be reconditioned as follows:

(1) On completion of the Life Test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring agency.

(2) After reworking, the contractor may submit the equipment for acceptance under the contract.

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4.5 Test Procedures - The procedures used for conducting Preproduction Tests, Acceptance Tests and Life Tests shall be prepared by the contractor and sent to the Bureau of Aeronautics for approval. The right is reserved by the Bureau of Aeronautics or the Naval Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.6 Pre-Submission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements. The design data submitted by the contractor for the preproduction model shall indicate its physical and electrical characteristics and other necessary test data which will establish that the equipment meets all the requirements of this specification.

4.7 Rejection and Retest - Electronic equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring agency.

4.8 Equipment Changes - After approval of the preproduction model, no changes shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This applies for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required the contractor shall also prepare an Electronic Material Change (EMC) and an Electronic Material Bulletin (EMB) to cover the approved change. ECP's, EMC's and EMB's shall be prepared in accordance with the requirements of SAR-300.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be packaged, packed and marked for shipment in accordance with the requirements of Specifications MIL-E-17555 and MIL-STD-129.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V, DC and 115 ± 1 V, AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

PATENT NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEM PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 15419

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